
DOCSNARY · DOCUMENT VALIDATION WRITE-UP

Docsnary: Document Intelligence & Validation

By Siddharth

WHAT I BUILT

I built Docsnary, an agentic document analysis and validation system for organizations that manage multiple related documents over time, such as vendor contracts and their amendments.

The problem is simple but messy: organizations often have several documents describing the same business reality, but the information changes between versions. Important facts such as vendor names, contract values, dates, and status can become inconsistent, while manually comparing every document is slow and error-prone.

Docsnary turns this into a structured workflow. A document is uploaded and associated with a case. The system processes it through visible workflow stages, extracts structured facts with supporting evidence, validates those facts against deterministic business rules, identifies changes and potential conflicts, and produces findings for human review. A reviewer can approve or reject individual findings before approved changes are committed to the cumulative case record.

When a newer document is processed against the same case, Docsnary can identify proposed changes against the information already committed. This allows the case to evolve over time while preserving the history and provenance of what changed and which document produced the change.

RESULTS

The current implementation provides a working end-to-end document workflow with a passing automated test suite.

The demonstrated vendor-contract workflow successfully:

- Uploads and processes synthetic vendor contract documents.
- Associates documents and runs with a case.
- Extracts structured fields including vendor, contract value, start date, end date, and status.
- Preserves supporting evidence for extracted facts.
- Validates extracted information using deterministic rules.
- Generates findings describing proposed changes.
- Allows a human reviewer to approve or reject findings individually.
- Commits approved findings into the cumulative case record.
- Prevents rejected findings from overwriting existing committed information.
- Produces a case report containing the current structured information.
- Supports subsequent document versions being processed against the same case.
- Maintains idempotent commit behavior, preventing already-committed findings from being committed again.
- Runs its automated tests without requiring a paid external AI API key by using the deterministic provider.

The system also exposes a machine-facing interface through MCP, allowing the workflow to be driven programmatically rather than requiring the React interface for every operation.

WHY THE DESIGN CHOICES MATTER

The most important design decision was to separate AI-assisted extraction from deterministic validation and persistence.

For local development and automated testing, Docsnary uses a deterministic provider that extracts facts from the declared document structure without requiring an external model or API key. This makes the core workflow reproducible and allows the test suite to verify behavior without spending money on API calls.

The architecture also supports external AI providers through the provider abstraction. When a real provider is configured, the AI can perform more flexible extraction from less-structured documents while the resulting structured facts continue through the same validation, review, and commit pipeline.

I deliberately keep important business rules deterministic. Required fields, evidence requirements, date validation, amount validation, commit behavior, and idempotency are handled by application code rather than asking an LLM to decide whether these guarantees are satisfied. This makes those behaviors predictable and testable.

The workflow is implemented using LangGraph so that processing is represented as explicit stages rather than a single model call hidden behind a UI. The workflow includes validation, retry, escalation, conflict handling, human review, and commit paths. This provides a clearer execution model and makes failure handling part of the system rather than an afterthought.

Human review remains a hard gate before changes become committed organizational data. An extracted value may be correct, incorrect, or contextually inappropriate, so the system does not silently turn model output into the source of truth.

PostgreSQL provides persistent state for documents, versions, runs, findings, evidence, and committed facts. Storage references are kept separately from database records, allowing the storage layer to evolve independently.

The result is a deliberate hybrid architecture: AI for flexible interpretation, deterministic code for guarantees, and humans for final authority.

HONEST LIMITATIONS

Docsnary is currently a focused engineering build rather than a production enterprise document platform.

The deterministic provider is intentionally limited in what document structures it can interpret. It is useful for reproducible development and testing, but it is not equivalent to a general-purpose language model. Less-structured documents require a configured external AI provider to provide flexible extraction.

The current demonstrated domain is vendor contracts and related documents. The architecture is designed to support other document domains, but each domain would require appropriate extraction schemas and validation rules.

The current implementation also does not attempt to provide every possible enterprise feature such as authentication and authorization, multi-tenant administration, production object storage, sophisticated OCR infrastructure, or a fully autonomous conflict-resolution system. Those would be natural production extensions rather than claims of functionality that the current build does not provide.

The system intentionally does not silently resolve contradictions. When information conflicts across documents, the design favors surfacing the disagreement and allowing a human to decide which information should become authoritative.

The build therefore prioritizes traceability, deterministic guarantees, human control, reproducibility, and incremental updates over pretending that one AI call can safely solve every document problem.

Docsnary's goal is not simply to extract information from documents. It is to turn a growing collection of changing documents into a reviewable, traceable, and continuously updated source of structured information.

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